

STA5228 Reliability and Life Data Project

Reliability Assessment of Dishwashers. A Multimodal Failure Analysis

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1. Introduction

In this course we learned how to do parametric and non-parametric reliability analysis on products or systems while taking in consideration only one failure mode. However, many products systems, components etc, may have more than one failure mode, where each failure mode is like a component in a series system. When one of the components fail the system fails. This project's concern is the reliability of dishwashers. Dishwashers are essential household appliances, and their reliability is paramount for user satisfaction. By investigating and modeling various failure scenarios, we seek to enhance our understanding of dishwasher performance and provide insights for potential improvements. The data set that we chose for the project contains two failure modes for the dishwashers. One failure mode is failure of the spray arm due to breakage and failure of the spray arm due to obstruction.

We are going to do a parametric and non-parametric analysis on the reliability of the dishwashers of our data set. First perform a parametric analysis taking into consideration each failure mode separately while considering the other failure mode times as right censored times. Next, we do a non-parametric analysis, giving Kaplan-Meier estimates of the survival probabilities for both cases aforementioned. Lastly we estimate the reliability taking in consideration both failure modes.

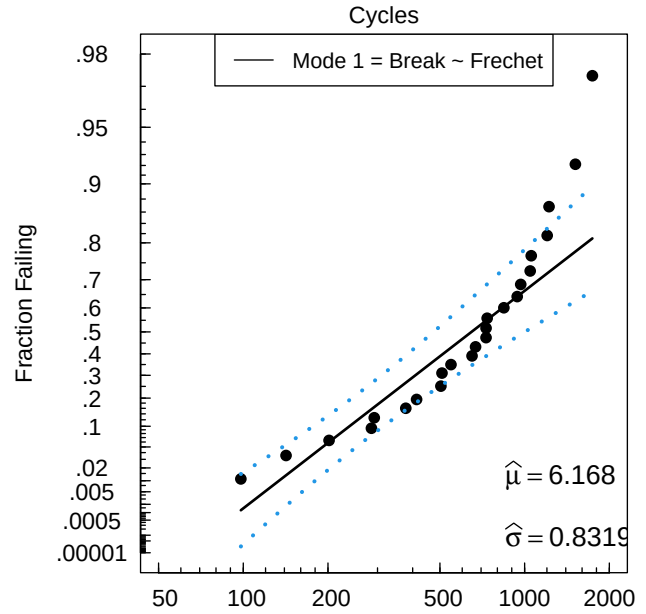
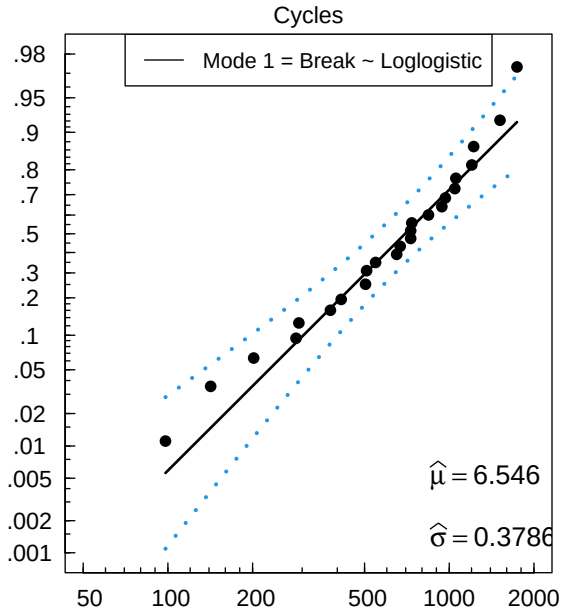
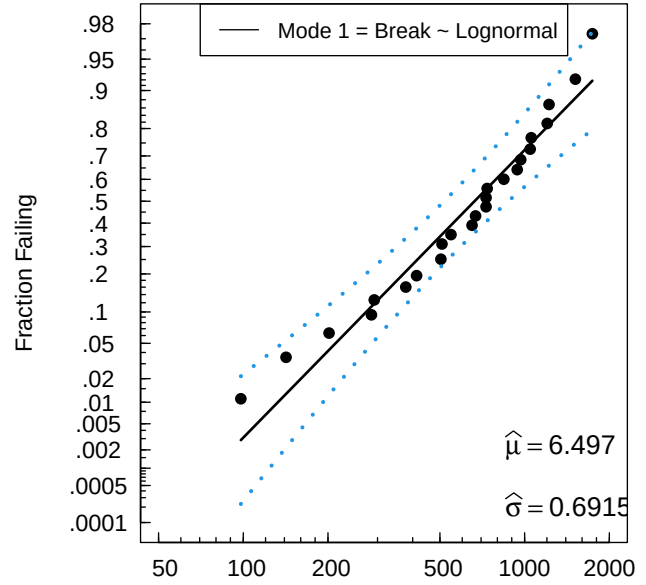
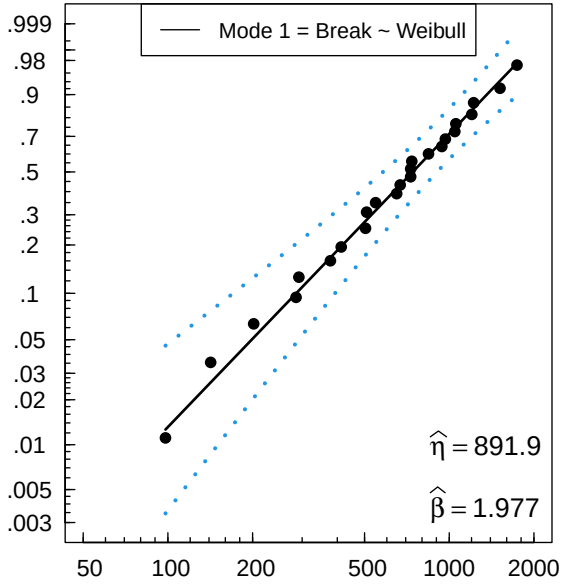
2. Parametric Analysis of the Dishwasher data

When failure times for the different failure modes are independent and the cause of failure is given for all units that have failed, we can analyze the individual failure modes separately. For this, the failures for failure mode j , are marked as failures and the failures for the other failure modes are marked as right-censored observations. This way, we can compute both, the parametric and non-parametric estimates of the marginal distribution for failure mode j .

2.1 Parametric Analysis for Failure Mode 1 = Break

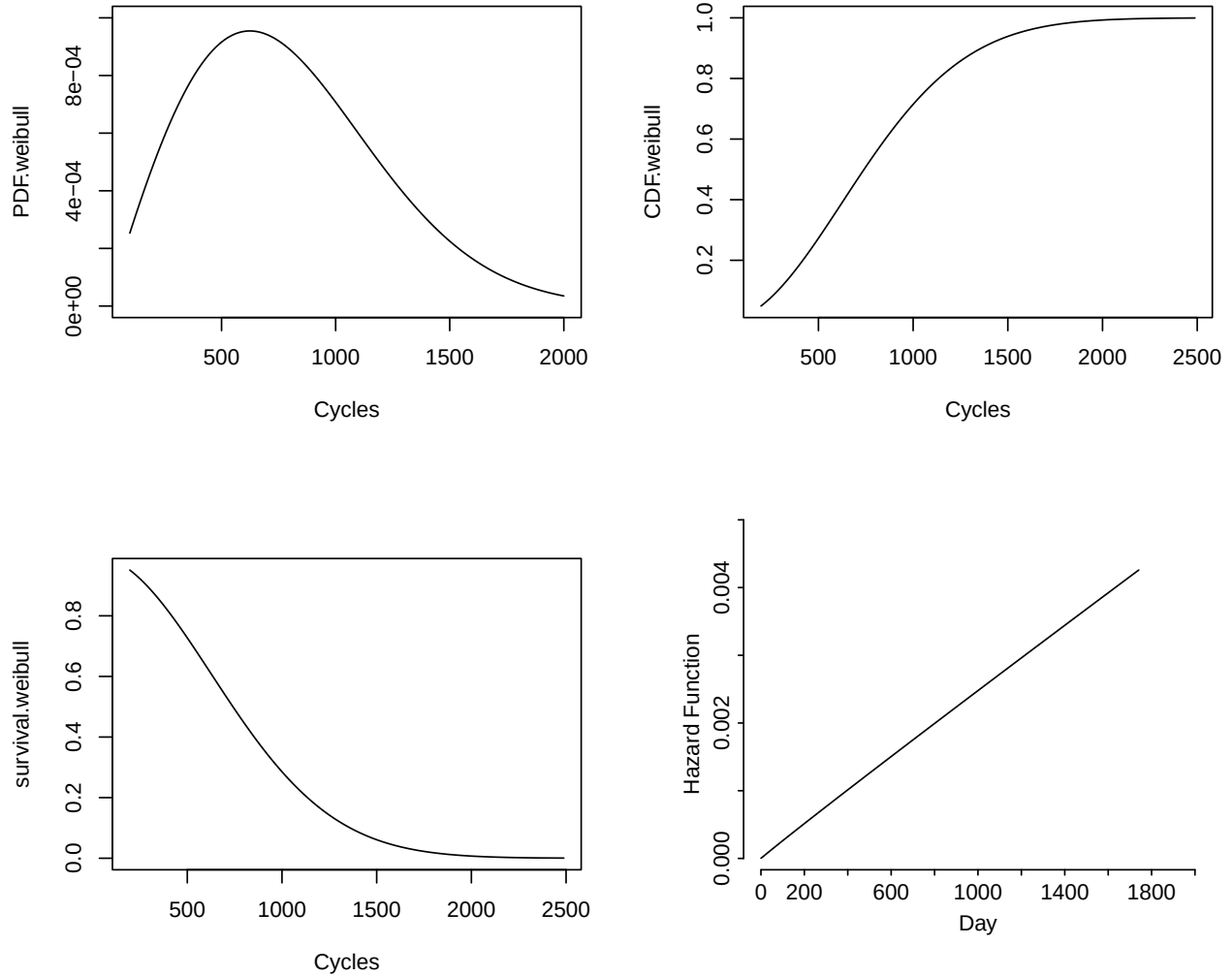
First we consider the failure mode break. This means that we are going to consider the failures for failure mode break as failures and the failures for failure mode obstruction are marked as right-censored observations.

In order to perform the parametric analysis for failure mode break, we need to find a distribution that fits our data. For this, we use probability plots. Below we have the probability plots for four different distributions.



From these plots we see that for this failure mode Weibull distribution is a better fit. Thus we are going to use this distribution for the parametric analysis.

In the next figure we have the pdf, CDF, survival and hazard function for the Weibull distribution with parameters, the ML estimates.



Under the assumption that the Weibull distribution is correct, we give the ML estimates for the reliability, quantiles and the hazard rate, as well as 95% confidence intervals for each of them. The ML estimate for the Mean Time To Failure is $MTTF = 790.6$ cycles. We can see that 10% of the dishwashers will fail due to arm spray break after about 87 cycles, and 90% of them will fail after about 1360 cycles. After 1000 cycles the probability that the spray arm will break is 0.7145, with a 95% confidence interval (0.5712, 0.8437).

The hazard rate is very small, which tells us that in general the spray arms take longer to break, after having completed many cycles.

```
## [1] "The maximum value of the log-likelihood function for the DishwasherBreak (assuming the Weibull
## [1] "The ML estimate of mean time to failure (MTTF) for the DishwasherBreak (assuming the Weibull di

##              MLE      Std.Err.   95% Lower   95% Upper
## mu           6.7933867  0.10183201  6.5937997  6.9929738
## sigma        0.5058884  0.07078502  0.3845499  0.6655133
## Weibull (eta) 891.9291745 90.82693913 730.5514499 1088.9549975
## Weibull (beta) 1.9767205 0.27658708  1.5025997  2.6004426

##      Cycles    Fhat Std.Err. 95% Lower 95% Upper
## [1,]    50 0.003355 0.002826 0.0006428 0.01741
## [2,]   100 0.013140 0.008603 0.0036280 0.04700
## [3,]   200 0.050730 0.023720 0.0201200 0.12490
## [4,]   500 0.272800 0.062770 0.1708000 0.41830
```

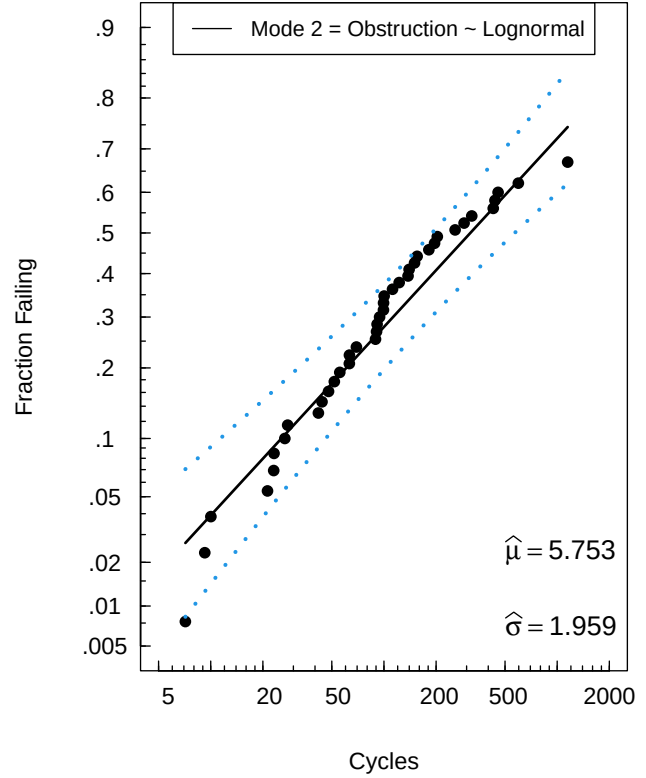
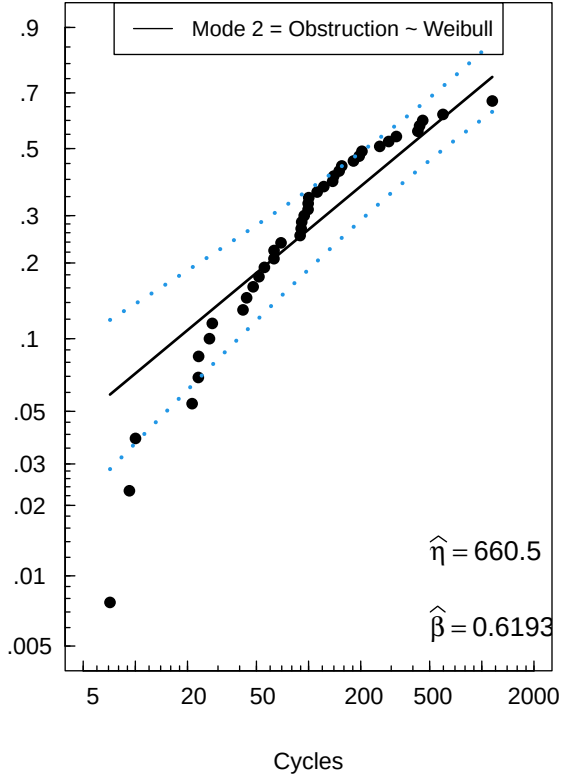
```
## [5,] 1000 0.714500 0.071640 0.5712000 0.84370
## [6,] 2000 0.992800 0.010060 0.9411000 0.99980
```

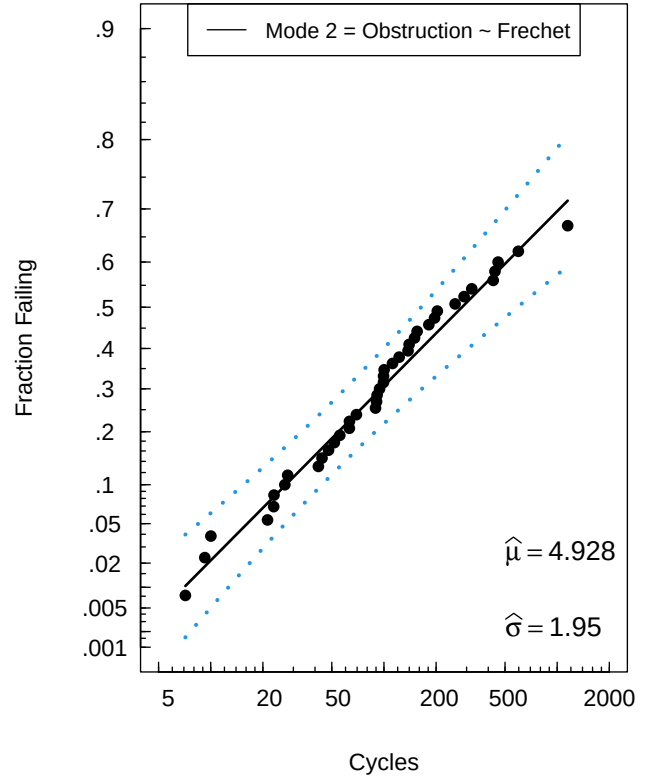
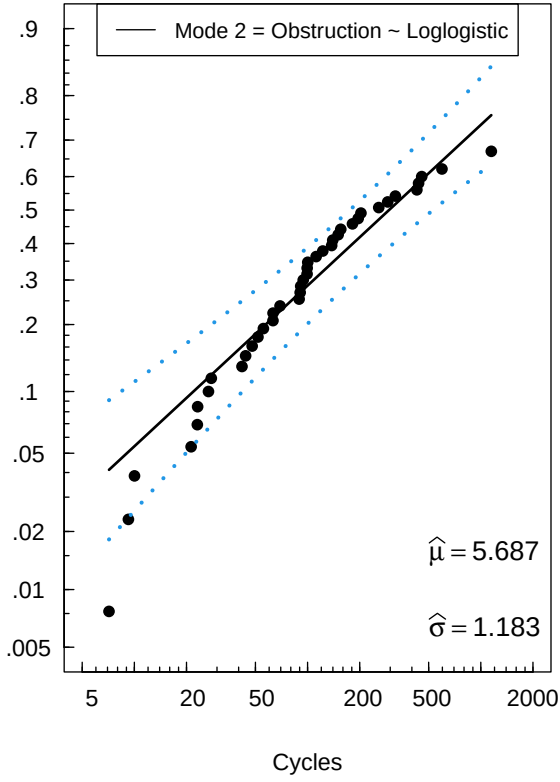
```
##           p Quanhhat Std.Err. 95% Lower 95% Upper
## [1,] 0.001 27.09 13.83 9.958 73.69
## [2,] 0.005 61.21 24.45 27.980 133.90
## [3,] 0.010 87.03 30.63 43.650 173.50
## [4,] 0.050 198.50 48.39 123.100 320.10
## [5,] 0.100 285.70 56.72 193.600 421.60
## [6,] 0.200 417.60 64.82 308.100 566.10
## [7,] 0.300 529.50 69.79 408.900 685.50
## [8,] 0.400 635.00 74.39 504.700 798.90
## [9,] 0.500 741.00 79.95 599.700 915.50
## [10,] 0.600 853.30 87.65 697.700 1044.00
## [11,] 0.700 979.70 99.14 803.500 1195.00
## [12,] 0.800 1135.00 117.50 926.200 1390.00
## [13,] 0.900 1360.00 152.00 1093.000 1693.00
## [14,] 0.990 1931.00 270.10 1468.000 2540.00
```

2.2 Parametric Analysis for Failure Mode 2 = Obstruction

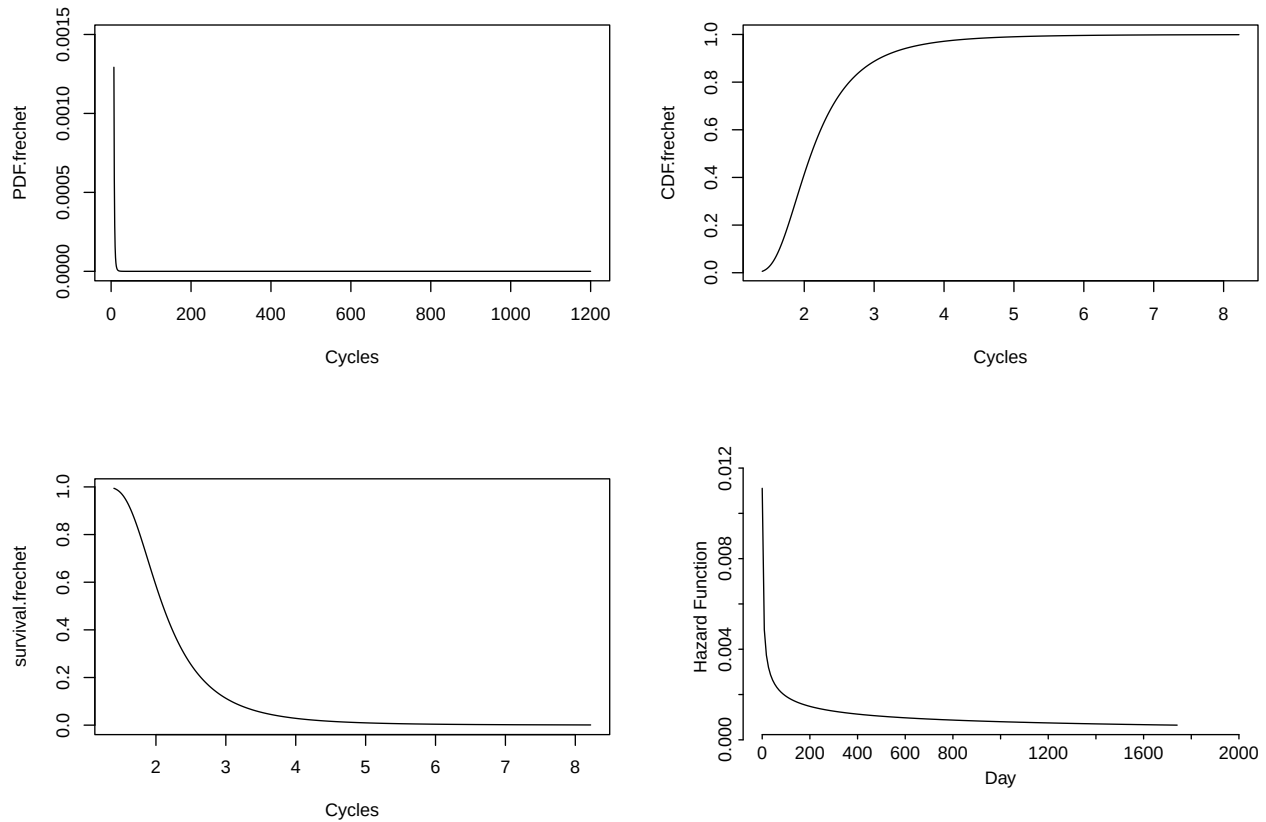
Now we consider the failure mode obstruction. This means that we are going to consider the failures for failure mode obstruction as failures and the failures for failure mode break are marked as right-censored observations.

Again, for the parametric analysis we need to find a distribution that fits well our data. Below we have the probability plots for four different distributions.





From these four plots, Frechet distributions seems to be a better fit for the failure mode obstruction. Assuming that this distribution is correct, we are going to perform the parametric analysis based on it. In the figure below we have the pdf, CDF, survival and hazard function for the Frechet distribution with parameters, the ML estimates.



Under the assumed Frechet distribution, we give the ML estimates for the reliability, quantiles and the hazard rate, as well as 95% confidence intervals for each of them. The ML estimate for the Mean Time To Failure under this distribution does not exist, since the shape parameter $\beta = \frac{1}{\sigma} = 0.513 < 1$.

We can see that 10% of the dishwashers will fail due to arm spray obstruction after about 7 cycles, and 90% of them will fail after about 11120 cycles. After 1000 cycles the probability that the spray arm will break is 0.6961, with a 95% confidence interval (0.5733, 0.7898).

The hazard rate estimates are in accordance with the graph above, as we can see that the values come decreasing, with the increase in the number of cycles.

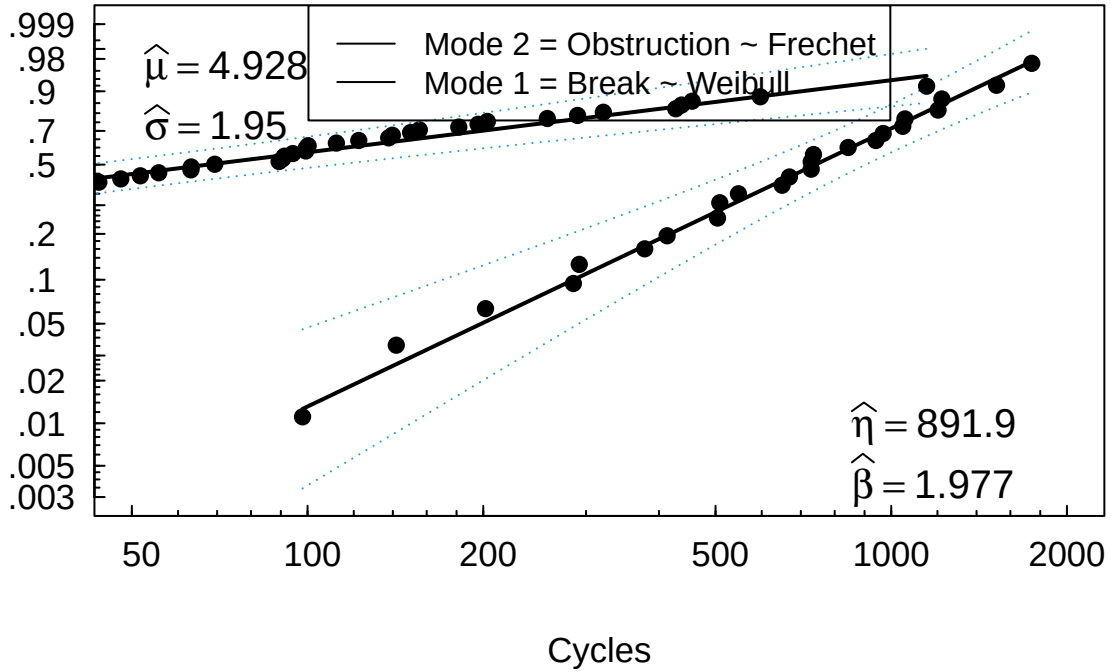
```
## [1] "The maximum value of the log-likelihood function for the DishwasherObstruction (assuming the f
## [1] "The ML estimate of mean time to failure (MTTF) for the DishwasherObstruction (assuming the frec
##           MLE Std.Err. 95% Lower 95% Upper
## mu      4.927572 0.2674624  4.403355  5.451789
## sigma  1.950379 0.2355861  1.539226  2.471358
##           Cycles      Fhat Std.Err. 95% Lower 95% Upper
## [1,]         5 0.004165 0.004423 0.0003314  0.02353
## [2,]        10 0.021460 0.013450 0.0050430  0.06143
## [3,]        20 0.067710 0.025410 0.0290600  0.12890
## [4,]        50 0.185800 0.039010 0.1165000  0.26770
## [5,]       100 0.307400 0.047170 0.2182000  0.40080
## [6,]       200 0.437400 0.053600 0.3310000  0.53880
## [7,]       500 0.596400 0.057070 0.4757000  0.69800
## [8,]      1000 0.696100 0.055180 0.5733000  0.78980
## [9,]      2000 0.775700 0.050280 0.6578000  0.85730
##           p  Quanhat Std.Err. 95% Lower 95% Upper
```

```

## [1,] 0.001 3.184e+00 1.341e+00      1.395 7.267e+00
## [2,] 0.005 5.342e+00 1.986e+00      2.578 1.107e+01
## [3,] 0.010 7.022e+00 2.440e+00      3.554 1.387e+01
## [4,] 0.050 1.624e+01 4.614e+00      9.309 2.834e+01
## [5,] 0.100 2.714e+01 6.988e+00     16.380 4.495e+01
## [6,] 0.200 5.457e+01 1.326e+01     33.890 8.787e+01
## [7,] 0.300 9.611e+01 2.426e+01     58.610 1.576e+02
## [8,] 0.400 1.637e+02 4.530e+01     95.180 2.816e+02
## [9,] 0.500 2.821e+02 8.844e+01    152.600 5.215e+02
## [10,] 0.600 5.117e+02 1.858e+02    251.100 1.043e+03
## [11,] 0.700 1.031e+03 4.431e+02    444.000 2.394e+03
## [12,] 0.800 2.574e+03 1.350e+03    920.400 7.196e+03
## [13,] 0.900 1.112e+04 7.632e+03   2898.000 4.268e+04
## [14,] 0.990 1.088e+06 1.327e+06  99600.000 1.188e+07

```

We construct a multiple probability plot that shows the parametric and non-parametric estimates for each of the failure modes together in one plot.



The plot shows the ML estimates for each failure mode and the non-parametric estimate of the proportion failing. Both distributions seem to fit each failure mode well.

3. Non-Parametric Analysis

3.1 Non-Parametric Analysis for Failure Mode 1 = Break

For the non-parametric analysis, we are going to compute the Kaplan-Meier estimates for the failing and survival probabilities.

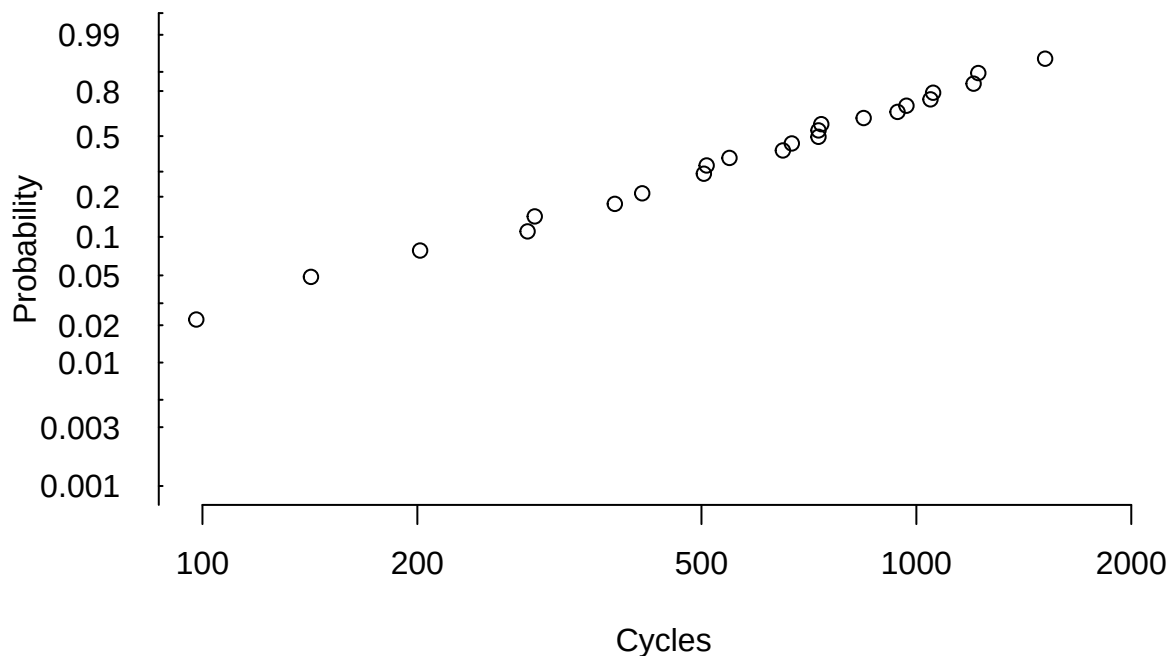
The table below gives these values. These values are close to what we got from the parametric analysis.

##	Failure	Cycles	S.hat	F.hat
## 21	Break	98.04	0.97777778	0.02222222
## 29	Break	141.90	0.95135135	0.04864865
## 34	Break	201.78	0.92162162	0.07837838

```
## 37 Break 285.38 0.88984157 0.11015843
## 39 Break 292.05 0.85688447 0.14311553
## 41 Break 378.10 0.82260909 0.17739091
## 42 Break 413.05 0.78833371 0.21166629
## 46 Break 503.88 0.70950034 0.29049966
## 47 Break 508.44 0.67008366 0.32991634
## 48 Break 547.25 0.63066697 0.36933303
## 50 Break 650.18 0.58862251 0.41137749
## 51 Break 669.18 0.54657804 0.45342196
## 52 Break 729.22 0.50453358 0.49546642
## 53 Break 729.30 0.46248911 0.53751089
## 54 Break 735.90 0.42044465 0.57955535
## 55 Break 843.60 0.37840018 0.62159982
## 56 Break 941.05 0.33635572 0.66364428
## 57 Break 968.55 0.29431125 0.70568875
## 58 Break 1046.52 0.25226679 0.74773321
## 59 Break 1055.45 0.21022232 0.78977768
## 61 Break 1202.70 0.15766674 0.84233326
## 62 Break 1221.00 0.10511116 0.89488884
## 63 Break 1514.70 0.05255558 0.94744442
## 64 Break 1740.75 0.00000000 1.00000000
```

We construct a probability plot for the failures of failure mode break.

Weibull Probability Scale



Using the survival package in R, we create a survival object in order to create a Kaplan-Meier survival curve and get the Kaplan-Meier estimates for the survival probabilities. The summary and the Kaplan-Meier curve are found below.

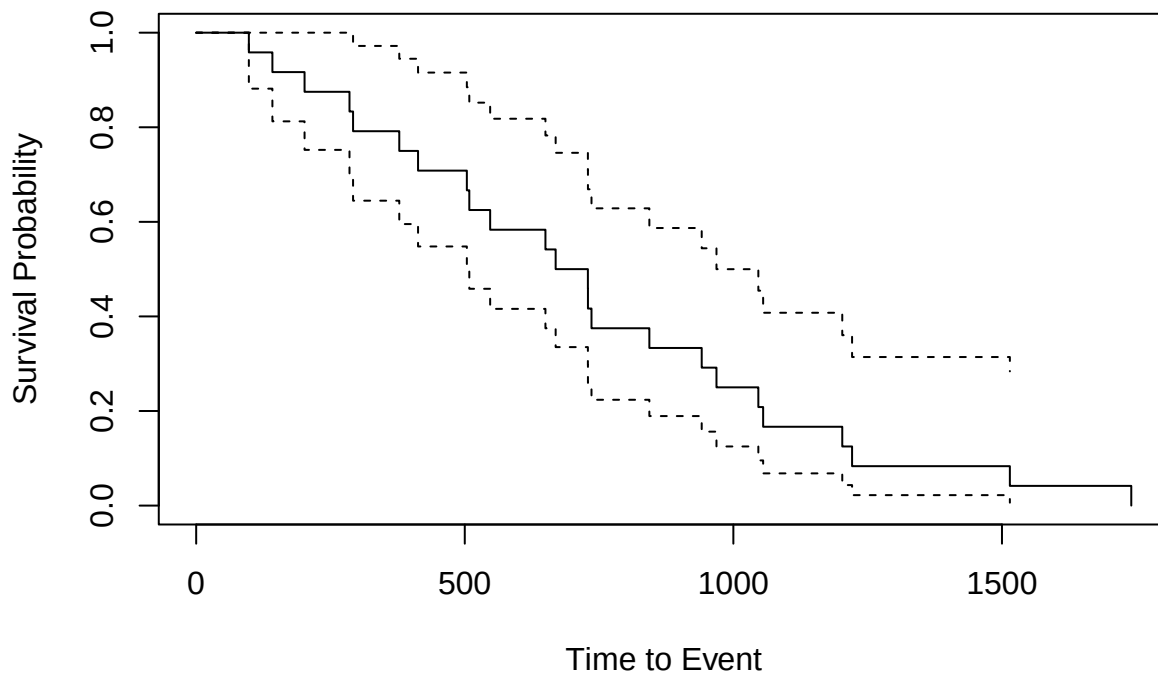
##	time	survival	n.risk	n.event	std.err	lower.95.CI	upper.95.CI
## 21	98.04	0.97727273	44	1	0.02246752	0.934214567	1.0000000
## 29	141.90	0.95012626	36	1	0.03454845	0.884769142	1.0000000

## 34	201.78	0.91947703	31	1	0.04502116	0.835339003	1.0000000
## 37	285.38	0.88663856	28	1	0.05407923	0.786735630	0.9992276
## 39	292.05	0.85253708	26	1	0.06182319	0.739583032	0.9827422
## 41	378.10	0.81701470	24	1	0.06869859	0.692877865	0.9633920
## 42	413.05	0.78149232	23	1	0.07433037	0.648580776	0.9416410
## 46	503.88	0.74036115	19	1	0.08100285	0.597466739	0.9174312
## 47	508.44	0.69922997	18	1	0.08631598	0.548964210	0.8906274
## 48	547.25	0.65809880	17	1	0.09050946	0.502601623	0.8617044
## 50	650.18	0.61422555	15	1	0.09451268	0.454308644	0.8304333
## 51	669.18	0.57035229	14	1	0.09741407	0.408094835	0.7971229
## 52	729.22	0.52647904	13	1	0.09931025	0.363762661	0.7619808
## 53	729.30	0.48260579	12	1	0.10025826	0.321189147	0.7251439
## 54	735.90	0.43873253	11	1	0.10028499	0.280306481	0.6866992
## 55	843.60	0.39485928	10	1	0.09939120	0.241092601	0.6466970
## 56	941.05	0.35098603	9	1	0.09755158	0.203568439	0.6051586
## 57	968.55	0.30711277	8	1	0.09471103	0.167800935	0.5620842
## 58	1046.52	0.26323952	7	1	0.09077565	0.133912871	0.5174637
## 59	1055.45	0.21936627	6	1	0.08559454	0.102103219	0.4713031
## 61	1202.70	0.16452470	4	1	0.07985495	0.063545604	0.4259677
## 62	1221.00	0.10968313	3	1	0.06956439	0.031643308	0.3801875
## 63	1514.70	0.05484157	2	1	0.05209222	0.008522778	0.3528893
## 64	1740.75	0.00000000	1	1	NaN	NA	NA

We can see that the Kaplan-Meier estimates for the survival probabilities found above match the results from the survival object.

The figure below gives us the Kaplan-Meier survival curve for failure mode break.

Kaplan-Meier Survival Curve



We see that as the number of cycles increases, the survival probability tends to zero, as expected.

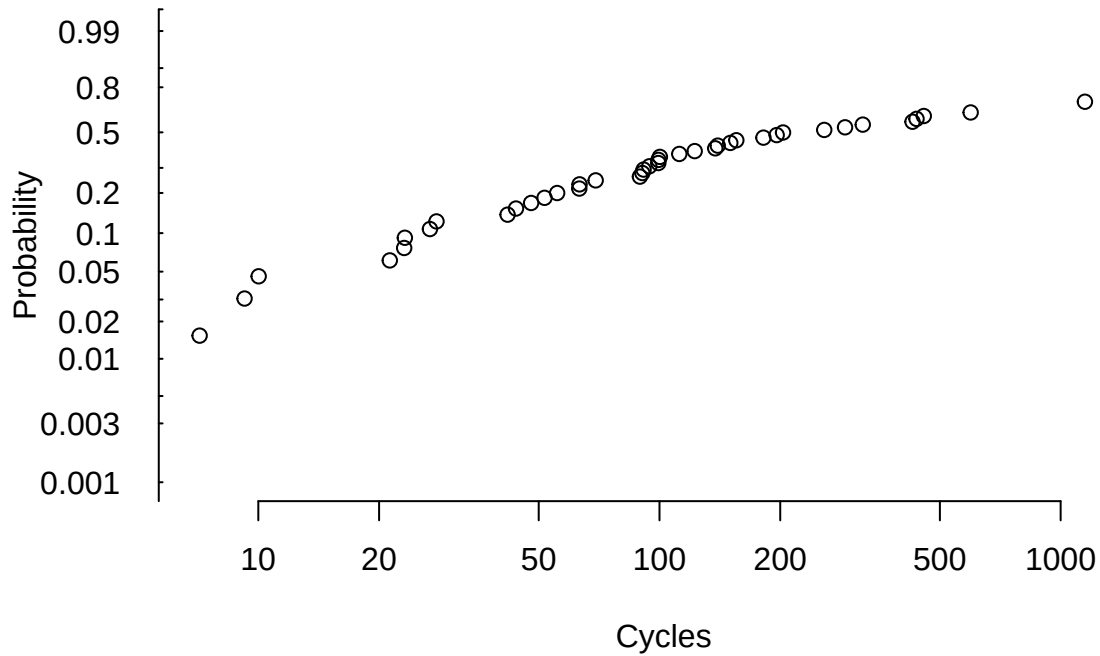
3.2 Non-Parametric Analysis for Failure Mode 2 = Obstruction

The non-parametric analysis for failure mode 2= obstruction. We repeat the same process as for failure mode 1 = break. First compute the Kaplan-Meier estimators, and construct a probability plot for obstruction failure times. Then we compare them with the values from the survival object for these failure times.

##	Failure	Cycles	S.hat1	F.hat1
## 1	Obstruction	7.14	0.9846154	0.01538462
## 2	Obstruction	9.24	0.9694675	0.03053254
## 3	Obstruction	10.02	0.9540791	0.04592092
## 4	Obstruction	21.27	0.9386907	0.06130929
## 5	Obstruction	23.10	0.9233023	0.07669766
## 6	Obstruction	23.19	0.9079140	0.09208603
## 7	Obstruction	26.78	0.8925256	0.10747441
## 8	Obstruction	27.81	0.8771372	0.12286278
## 9	Obstruction	41.82	0.8617488	0.13825115
## 10	Obstruction	43.89	0.8463605	0.15363952
## 11	Obstruction	47.87	0.8309721	0.16902790
## 12	Obstruction	51.70	0.8155837	0.18441627
## 13	Obstruction	55.59	0.8001954	0.19980464
## 14	Obstruction	63.12	0.7848070	0.21519301
## 15	Obstruction	63.20	0.7694186	0.23058138
## 16	Obstruction	69.34	0.7540302	0.24596976
## 17	Obstruction	89.38	0.7386419	0.26135813
## 18	Obstruction	90.63	0.7232535	0.27674650
## 19	Obstruction	91.28	0.7078651	0.29213487
## 20	Obstruction	94.35	0.6924768	0.30752325
## 22	Obstruction	99.33	0.6767386	0.32326135
## 23	Obstruction	99.44	0.6610005	0.33899946
## 24	Obstruction	100.23	0.6452624	0.35473757
## 25	Obstruction	112.04	0.6295243	0.37047568
## 26	Obstruction	122.40	0.6137862	0.38621379
## 27	Obstruction	137.72	0.5980481	0.40195189
## 28	Obstruction	139.72	0.5823100	0.41769000
## 30	Obstruction	150.15	0.5661347	0.43386528
## 31	Obstruction	155.43	0.5499594	0.45004056
## 32	Obstruction	181.60	0.5337842	0.46621584
## 33	Obstruction	195.94	0.5176089	0.48239111
## 35	Obstruction	203.22	0.5009118	0.49908817
## 36	Obstruction	257.47	0.4842148	0.51578523
## 38	Obstruction	290.11	0.4669214	0.53307862
## 40	Obstruction	321.20	0.4489629	0.55103713
## 43	Obstruction	427.35	0.4294427	0.57055726
## 44	Obstruction	437.29	0.4099226	0.59007738
## 45	Obstruction	455.87	0.3904025	0.60959751
## 49	Obstruction	596.67	0.3660023	0.63399766
## 60	Obstruction	1149.66	0.2928019	0.70719813

Probability plot for the failures of failure mode obstruction.

Weibull Probability Scale



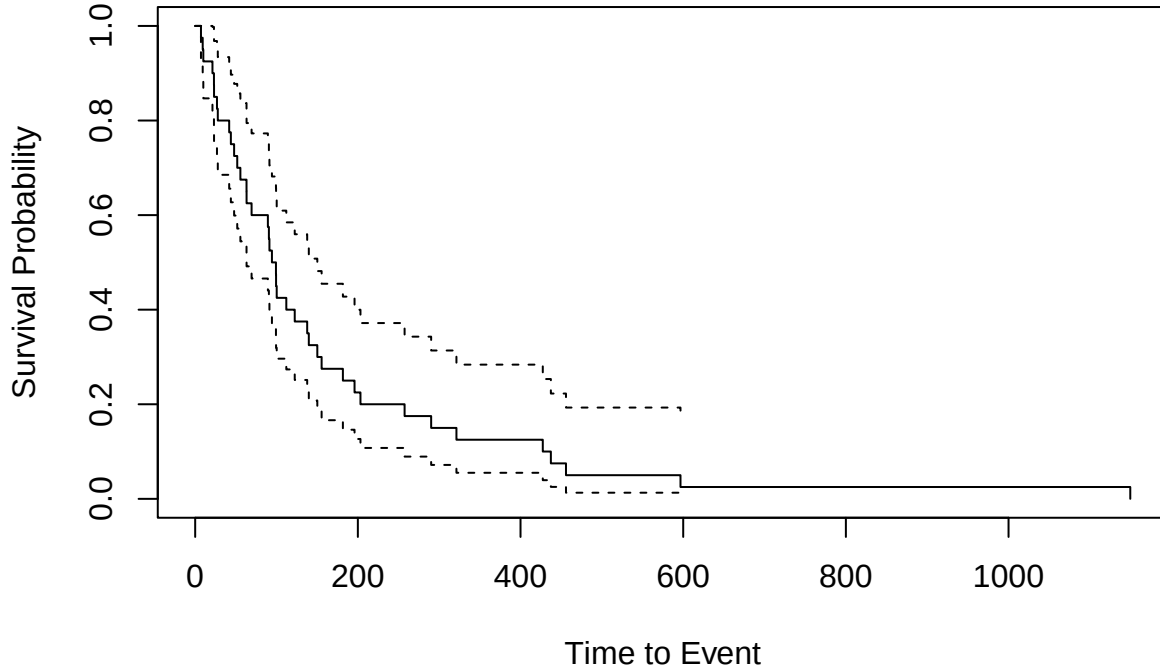
The survival probabilities, standard error and 95% confidence interval for them with failure mode obstruction.

##	time	survival	n.risk	n.event	std.err	lower.95.CI	upper.95.CI
## 1	7.14	0.9843750	64	1	0.01550245	0.9544549	1.0000000
## 2	9.24	0.9687500	63	1	0.02174908	0.9270468	1.0000000
## 3	10.02	0.9531250	62	1	0.02642139	0.9027217	1.0000000
## 4	21.27	0.9375000	61	1	0.03025768	0.8800328	0.9987199
## 5	23.10	0.9218750	60	1	0.03354602	0.8584159	0.9900254
## 6	23.19	0.9062500	59	1	0.03643507	0.8375797	0.9805504
## 7	26.78	0.8906250	58	1	0.03901364	0.8173502	0.9704688
## 8	27.81	0.8750000	57	1	0.04133986	0.7976136	0.9598946
## 9	41.82	0.8593750	56	1	0.04345429	0.7782905	0.9489071
## 10	43.89	0.8437500	55	1	0.04538652	0.7593228	0.9375645
## 11	47.87	0.8281250	54	1	0.04715897	0.7406667	0.9259105
## 12	51.70	0.8125000	53	1	0.04878905	0.7222880	0.9139793
## 13	55.59	0.7968750	52	1	0.05029060	0.7041595	0.9017981
## 14	63.12	0.7812500	51	1	0.05167483	0.6862595	0.8893889
## 15	63.20	0.7656250	50	1	0.05295094	0.6685696	0.8767698
## 16	69.34	0.7500000	49	1	0.05412659	0.6510751	0.8639556
## 17	89.38	0.7343750	48	1	0.05520818	0.6337632	0.8509592
## 18	90.63	0.7187500	47	1	0.05620115	0.6166233	0.8377912
## 19	91.28	0.7031250	46	1	0.05711012	0.5996463	0.8244606
## 20	94.35	0.6875000	45	1	0.05793905	0.5828244	0.8109754
## 22	99.33	0.6715116	43	1	0.05875624	0.5656851	0.7971359
## 23	99.44	0.6555233	42	1	0.05949285	0.5487016	0.7831411
## 24	100.23	0.6395349	41	1	0.06015185	0.5318681	0.7689968
## 25	112.04	0.6235465	40	1	0.06073575	0.5151796	0.7547082
## 26	122.40	0.6075581	39	1	0.06124671	0.4986318	0.7402795
## 27	137.72	0.5915698	38	1	0.06168653	0.4822209	0.7257146
## 28	139.72	0.5755814	37	1	0.06205674	0.4659441	0.7110165

## 30	150.15	0.5591362	35	1	0.06242468	0.4492472	0.6959049
## 31	155.43	0.5426910	34	1	0.06271742	0.4326946	0.6806500
## 32	181.60	0.5262458	33	1	0.06293603	0.4162841	0.6652541
## 33	195.94	0.5098007	32	1	0.06308126	0.4000140	0.6497191
## 35	203.22	0.4928073	30	1	0.06322604	0.3832392	0.6337009
## 36	257.47	0.4758140	29	1	0.06328831	0.3666218	0.6175271
## 38	290.11	0.4581912	27	1	0.06335034	0.3494283	0.6008076
## 40	321.20	0.4398636	25	1	0.06341209	0.3315936	0.5834852
## 43	427.35	0.4198698	22	1	0.06360368	0.3120117	0.5650128
## 44	437.29	0.3998760	21	1	0.06363992	0.2927233	0.5462523
## 45	455.87	0.3798822	20	1	0.06352105	0.2737268	0.5272061
## 49	596.67	0.3561395	16	1	0.06383417	0.2506406	0.5060447
## 60	1149.66	0.2849116	5	1	0.08164928	0.1624707	0.4996265

The survival probabilities of the survival object agree with the Kaplan-Meier estimates that we found above. The Kaplan-Meier survival curve for this case is given below.

Kaplan–Meier Survival Curve



4. Consider both Failure Modes

Now we are going to take into consideration both failure modes of the system. For this we are going to refer to Chapter 16 of Statistical Methods for Reliability Data (William Q. Meeker 2021). Let T_1 and T_2 be the failure times for failure mode 1 and failure mode 2 respectively. Then, the joint distribution of T_1 and T_2 is $F(t_1, t_2) = \Pr(T_1 \leq t_1, T_2 \leq t_2)$ and it completely describes the multiple failure modes probability model. \

The marginal distribution for mode j is $F_j(t) = \Pr(T_j \leq t)$. We can interpret this as the cdf for mode j if only mode j is active, where $j = 1, 2$ in our case. Previously we made the strong assumption that the two failure modes are independent of each other. In that case the joint distribution is

$$F(t_1, t_2) = \prod_{j=1}^J F_j(t_j)$$

The system failure time is the first failure time. This means that the other failure times for the other modes are unobserved. So,

$$T = \min T_1, T_2$$

Becasue we assumed the failure modes to be independent, then the cdf for the random variable T is given by ,

$$F_T(t) = \Pr(T \leq t) = 1 - \Pr(T > t) = 1 - \Pr(T_1 > t_1 \cap T_2 > t_2)$$

$$F_T(t) = 1 - \prod_{j=1}^2 [1 - F_j(t)]$$

For failure mode 1 = break we assumed that the failures followed a Weibull distribution, and for failure mode 2 = obstruction, we assumed a Frechet distribution.

So, then we have,

$$\text{the marginal cdf of first falure mode: } F_1(t) = 1 - \exp \left[- \left(\frac{t}{\eta_1} \right)^{\beta_1} \right]$$

$$\text{and the marginal cdf of second failure mode: } \exp \left[- \left(\frac{\eta_2}{t} \right)^{\beta_2} \right].$$

Substituting above we get that the system's cdf is

$$F_T(t) = 1 - \exp \left[- \left(\frac{t}{\eta_1} \right)^{\beta_1} \right] + \exp \left[- \left(\frac{t}{\eta_1} \right)^{\beta_1} - \left(\frac{t}{\eta_2} \right)^{-\beta_2} \right]$$

where, $\eta_2 = \exp(\mu)$ and $\beta_2 = \frac{1}{\sigma}$.

The quantiles of the system failure-time distribution can be obtained numerically by solving for t_p in $F_T(t_p) = p$.

Similarly, the MTTF is

$$E(T) = \int_0^\infty [1 - F_T(t)] dt = \int_0^\infty \exp \left[- \left(\frac{t}{\eta_1} \right)^{\beta_1} \right] + \exp \left[- \left(\frac{t}{\eta_1} \right)^{\beta_1} - \left(\frac{t}{\eta_2} \right)^{\beta_2} \right] dt$$

An estimate of the failure-time cdf for the system with all failure modes active is obtained by substituting the ML estimates for the parameters for all of the failure modes into the system cdf.

Since we already obtained the ML estimates for all the failure modes, we can estimate the cdf of the system failure-time. So we will have the following table, showing the failure and the survival probability for each system failure-time.

##		Cycles	F.hat.joint	S.hat.joint
##	[1,]	7.14	0.01044737	0.989552626
##	[2,]	9.24	0.01838823	0.981611771
##	[3,]	10.02	0.02164128	0.978358716
##	[4,]	21.27	0.07411156	0.925888445
##	[5,]	23.10	0.08260727	0.917392726

##	[6,]	23.19	0.08302153	0.916978472
##	[7,]	26.78	0.09923748	0.900762521
##	[8,]	27.81	0.10377189	0.896228109
##	[9,]	41.82	0.15996571	0.840034285
##	[10,]	43.89	0.16744264	0.832557358
##	[11,]	47.87	0.18128022	0.818719779
##	[12,]	51.70	0.19396779	0.806032209
##	[13,]	55.59	0.20626959	0.793730413
##	[14,]	63.12	0.22856942	0.771430584
##	[15,]	63.20	0.22879642	0.771203585
##	[16,]	69.34	0.24565056	0.754349440
##	[17,]	89.38	0.29402454	0.705975459
##	[18,]	90.63	0.29675791	0.703242090
##	[19,]	91.28	0.29816773	0.701832270
##	[20,]	94.35	0.30472275	0.695277249
##	[21,]	98.04	0.31238435	0.687615649
##	[22,]	99.33	0.31500930	0.684990700
##	[23,]	99.44	0.31523189	0.684768114
##	[24,]	100.23	0.31682477	0.683175232
##	[25,]	112.04	0.33952352	0.660476483
##	[26,]	122.40	0.35790601	0.642093993
##	[27,]	137.72	0.38290755	0.617092453
##	[28,]	139.72	0.38600270	0.613997296
##	[29,]	141.90	0.38933578	0.610664218
##	[30,]	150.15	0.40158555	0.598414454
##	[31,]	155.43	0.40914251	0.590857493
##	[32,]	181.60	0.44381066	0.556189345
##	[33,]	195.94	0.46114915	0.538850850
##	[34,]	201.78	0.46792588	0.532074124
##	[35,]	203.22	0.46957335	0.530426655
##	[36,]	257.47	0.52600880	0.473991200
##	[37,]	285.38	0.55166001	0.448339986
##	[38,]	290.11	0.55582680	0.444173205
##	[39,]	292.05	0.55752187	0.442478132
##	[40,]	321.20	0.58208907	0.417910927
##	[41,]	378.10	0.62592747	0.374072533
##	[42,]	413.05	0.65065937	0.349340630
##	[43,]	427.35	0.66036561	0.339634390
##	[44,]	437.29	0.66698094	0.333019058
##	[45,]	455.87	0.67907004	0.320929959
##	[46,]	503.88	0.70875330	0.291246705
##	[47,]	508.44	0.71146259	0.288537406
##	[48,]	547.25	0.73379488	0.266205116
##	[49,]	596.67	0.76044005	0.239559951
##	[50,]	650.18	0.78715607	0.212843926
##	[51,]	669.18	0.79612919	0.203870814
##	[52,]	729.22	0.82277719	0.177222812
##	[53,]	729.30	0.82281100	0.177189003
##	[54,]	735.90	0.82558489	0.174415110
##	[55,]	843.60	0.86667581	0.133324190
##	[56,]	941.05	0.89741383	0.102586170
##	[57,]	968.55	0.90505009	0.094949914
##	[58,]	1046.52	0.92438212	0.075617882
##	[59,]	1055.45	0.92638693	0.073613068

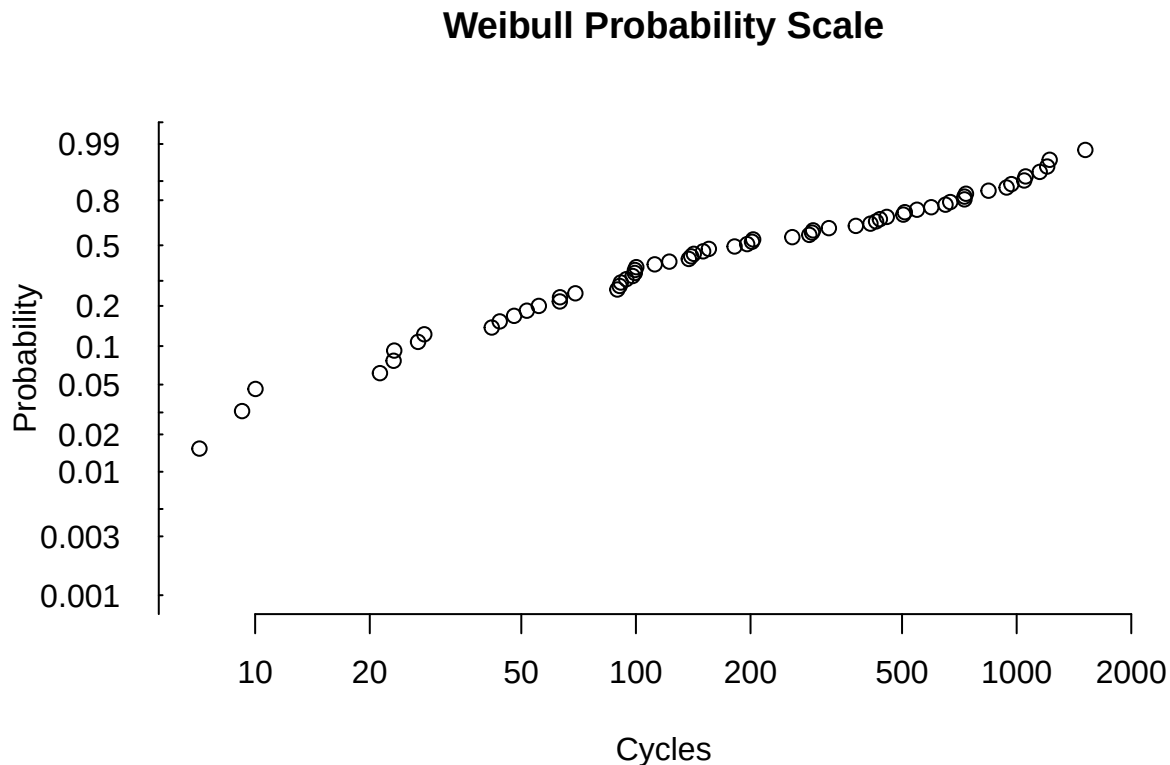
```
## [60,] 1149.66 0.94511663 0.054883366
## [61,] 1202.70 0.95386076 0.046139241
## [62,] 1221.00 0.95660288 0.043397122
## [63,] 1514.70 0.98530551 0.014694490
## [64,] 1740.75 0.99439377 0.005606226
```

We can also compute the Kaplan-Meier estimate for the system failure-times. These estimates are,

##		Cycles	F.hat.KM	S.hat.KM
##	[1,]	7.14	0.01538462	0.98461538
##	[2,]	9.24	0.03076923	0.96923077
##	[3,]	10.02	0.04615385	0.95384615
##	[4,]	21.27	0.06153846	0.93846154
##	[5,]	23.10	0.07692308	0.92307692
##	[6,]	23.19	0.09230769	0.90769231
##	[7,]	26.78	0.10769231	0.89230769
##	[8,]	27.81	0.12307692	0.87692308
##	[9,]	41.82	0.13846154	0.86153846
##	[10,]	43.89	0.15384615	0.84615385
##	[11,]	47.87	0.16923077	0.83076923
##	[12,]	51.70	0.18461538	0.81538462
##	[13,]	55.59	0.20000000	0.80000000
##	[14,]	63.12	0.21538462	0.78461538
##	[15,]	63.20	0.23076923	0.76923077
##	[16,]	69.34	0.24615385	0.75384615
##	[17,]	89.38	0.26153846	0.73846154
##	[18,]	90.63	0.27692308	0.72307692
##	[19,]	91.28	0.29230769	0.70769231
##	[20,]	94.35	0.30769231	0.69230769
##	[21,]	98.04	0.32307692	0.67692308
##	[22,]	99.33	0.33846154	0.66153846
##	[23,]	99.44	0.35384615	0.64615385
##	[24,]	100.23	0.36923077	0.63076923
##	[25,]	112.04	0.38461538	0.61538462
##	[26,]	122.40	0.40000000	0.60000000
##	[27,]	137.72	0.41538462	0.58461538
##	[28,]	139.72	0.43076923	0.56923077
##	[29,]	141.90	0.44615385	0.55384615
##	[30,]	150.15	0.46153846	0.53846154
##	[31,]	155.43	0.47692308	0.52307692
##	[32,]	181.60	0.49230769	0.50769231
##	[33,]	195.94	0.50769231	0.49230769
##	[34,]	201.78	0.52307692	0.47692308
##	[35,]	203.22	0.53846154	0.46153846
##	[36,]	257.47	0.55384615	0.44615385
##	[37,]	285.38	0.56923077	0.43076923
##	[38,]	290.11	0.58461538	0.41538462
##	[39,]	292.05	0.60000000	0.40000000
##	[40,]	321.20	0.61538462	0.38461538
##	[41,]	378.10	0.63076923	0.36923077
##	[42,]	413.05	0.64615385	0.35384615
##	[43,]	427.35	0.66153846	0.33846154
##	[44,]	437.29	0.67692308	0.32307692
##	[45,]	455.87	0.69230769	0.30769231
##	[46,]	503.88	0.70769231	0.29230769
##	[47,]	508.44	0.72393162	0.27606838
##	[48,]	547.25	0.74017094	0.25982906
##	[49,]	596.67	0.75641026	0.24358974
##	[50,]	650.18	0.77264957	0.22735043
##	[51,]	669.18	0.78888889	0.21111111
##	[52,]	729.22	0.80512821	0.19487179
##	[53,]	729.30	0.82136752	0.17863248

```
## [54,] 735.90 0.83760684 0.16239316
## [55,] 843.60 0.85384615 0.14615385
## [56,] 941.05 0.87008547 0.12991453
## [57,] 968.55 0.88632479 0.11367521
## [58,] 1046.52 0.90256410 0.09743590
## [59,] 1055.45 0.91880342 0.08119658
## [60,] 1149.66 0.93504274 0.06495726
## [61,] 1202.70 0.95128205 0.04871795
## [62,] 1221.00 0.96752137 0.03247863
## [63,] 1514.70 0.98376068 0.01623932
## [64,] 1740.75 1.00000000 0.00000000
```

Lastly, we plot the Kaplan-Meier estimate of the cdf versus cycles in a Weibull Probability Plot Scale.



5. Comparing the two failure mode distributions

Previously we assumed that each of the two failure modes followed two different distributions, Weibull and Frechet, respectively.

Now, let's suppose that also the second failure mode, obstruction, follows a Weibull Distribution. We can perform a Likelihood ratio test, to check if under this new assumption, the two distributions would have the same shape parameter.

Again, we create a survival object and fit two separate Weibull distributions to the two marginal distributions. So, we have,

```
## [[1]]
## Call:
## survreg(formula = Surv(time = Cycles, time2 = Cycles, type = "interval2") ~
##       1, data = subset(Dishwasher, Dishwasher$Failure == failure),
##       dist = "weibull")
```

```
##
## Coefficients:
## (Intercept)
##      6.692854
##
## Scale= 0.5553359
##
## Loglik(model)= -184.1   Loglik(intercept only)= -184.1
## n= 25
##
## [[2]]
## Call:
## survreg(formula = Surv(time = Cycles, time2 = Cycles, type = "interval2") ~
##      1, data = subset(Dishwasher, Dishwasher$Failure == failure),
##      dist = "weibull")
##
## Coefficients:
## (Intercept)
##      5.064521
##
## Scale= 1.076193
##
## Loglik(model)= -243.9   Loglik(intercept only)= -243.9
## n= 40
## [1] -428
```

For the marginal distribution of failure mode break, we have that the maximum log-likelihood is -184.1, and for the marginal distribution of failure mode obstruction the maximum log-likelihood is -243.9. The sum of the two equals -428.

We can also compute the log-likelihood of each of the separate distributions independently. The likelihood function for all exact failure times based on a Weibull distribution with pdf the following

$$f(t; \eta, \beta) = \frac{\beta}{\eta} \left(\frac{t}{\eta}\right)^{\beta-1} \exp \left[- \left(\frac{t}{\eta}\right)^{\beta} \right]$$

is going to be

$$\begin{aligned} L(\eta, \beta; t) &= \prod_{i=1}^n \frac{\beta}{\eta} \left(\frac{t_i}{\eta}\right)^{\beta-1} \exp \left[- \left(\frac{t_i}{\eta}\right)^{\beta} \right] \\ &= \left(\frac{\beta}{\eta^{\beta}}\right)^n \prod_{i=1}^n t_i^{\beta-1} \exp \left(- \frac{1}{\eta^{\beta}} \sum_{i=1}^n t_i^{\beta} \right) \end{aligned}$$

and the log-likelihood is given by

$$\mathcal{L}(\eta, \beta; t) = n \ln(\beta) - n\beta \ln(\eta) + (\beta - 1) \sum_{i=1}^n \ln(t_i) - \frac{1}{\eta^{\beta}} \sum_{i=1}^n t_i^{\beta}$$

Using `optim` function we get the following results

```

## $par
##      shape      scale
##  1.800108 807.163611
##
## $value
## [1] 184.0823
##
## $counts
## function gradient
##      117      NA
##
## $convergence
## [1] 0
##
## $message
## NULL

##      shape      scale
##  1.800108 807.163611

## $par
##      shape      scale
##  0.9299902 158.3908786
##
## $value
## [1] 243.9283
##
## $counts
## function gradient
##      101      NA
##
## $convergence
## [1] 0
##
## $message
## NULL

##      shape      scale
##  0.9299902 158.3908786

```

Computing the log-likelihood independently for each distribution gives the same results as the survival function.

Next we fit distributions with the same shape parameter for both.

```

## Call:
## survreg(formula = surv_object3 ~ as.factor(Failure), data = Dishwasher,
##      dist = "weibull")
##
## Coefficients:
##              (Intercept) as.factor(Failure)Obstruction
##              6.590515              -1.436178
##
## Scale= 0.9138773
##
## Loglik(model)= -433.1   Loglik(intercept only)= -449.7
##  Chisq= 33.22 on 1 degrees of freedom, p= 8.23e-09

```

```
## n= 65
## [1] 10.2
## [1] 3.841459
```

Equal sigma distributions gives a maximum likelihood equal to -433.1.

Then the $LR_{test} = 10.2$, whereas $chisq(0.95, 1 = 1.841)$. Since $LR_{tests} > chisq(0.95, 1 = 1.841)$, then we reject the hypothesis that both distributions have equal shape parameters. Therefore, the two failure modes have statistically significant differences between each-other.

6. Conclusions

On conclusion we can say that the Mean Time To Failure of spray arms due to breaks was 790.6 cycles. For obstructions the MTTF did not exist.

Referring to the Kaplan-Meier estimator of the survival probabilities for each of the three scenarios we found that:

About 95% of the spray arms survived breaks for at least 141.9 cycles, whereas 95% of spray arms survived obstruction for at least 10.2 cycles. About 95% of spray arms survived both breaks and obstructions for at least 10.2 cycles. From our results it seems that the obstruction failure mode needs more attention from the dishwasher engineers. They should focus on finding solutions on how to make the spray arm more resistant to obstructions.

References

William Q. Meeker, Francis G. Pascual, Luis A. Escobar. 2021. *Statistical Methods for Reliability Data, 2nd Edition*. Wiley.

Appendix

```
knitr::opts_chunk$set(echo = FALSE)
library(SMRD)
# Compatibility patch for SMRD on newer versions of R
CheckPrintDataName_fixed <- function() {
  the.par <- par(xpd = TRUE)
  on.exit(par(the.par))

  if (SMRDOptions("SMRD.DateOnPlot")) {
    text(x.loc(1), y.loc(-0.2), date(), adj = 1, cex = 0.8)
  }

  text(
    x.loc(1),
    y.loc(-0.23),
    GetSMRDDefault("SMRD.NameOnPlot"),
    adj = 1,
    cex = 0.8
  )
}

environment(CheckPrintDataName_fixed) <- asNamespace("SMRD")

assignInNamespace(
  "CheckPrintDataName",
  CheckPrintDataName_fixed,
  ns = "SMRD"
)
library(MASS)
library(survival)
library(survminer)
library(ComparisonSurv)
library(muhaz)
library(rms)
library(VGAM)
library(cmprsk)

DishwasherBreak <- read.csv("DishwasherReliabilityBreak.csv")

## Failure Mode = Break, Censoring = Obstruction

## Weibull fits better

DishwasherBreak.ld <- frame.to.ld(DishwasherBreak,
  response.column = 2,
  censor.column = 3,
```

```

time.units = "Cycles")

par(mfrow = c(1,2))
mleprobplot(DishwasherBreak.ld, distribution="Weibull")
legend("top", legend = "Mode 1 = Break ~ Weibull" , lty = 1:1)

mleprobplot(DishwasherBreak.ld, distribution="lognormal")
legend("top", legend = "Mode 1 = Break ~ Lognormal" , lty = 1:1)

par(mfrow = c(1,1))

par(mfrow = c(1,2))
mleprobplot(DishwasherBreak.ld, distribution="loglogistic")
legend("top", legend = "Mode 1 = Break ~ Loglogistic" , lty = 1:1)

mleprobplot(DishwasherBreak.ld, distribution="frechet")
legend("top", legend = "Mode 1 = Break ~ Frechet" , lty = 1:1)
par(mfrow = c(1,1))

x<-98:2000

PDF.weibull <- dweibull(x, shape = 1.977, scale = 891.9)

q <- qweibull(x/2000,shape = 1.977, scale = 891.9)

CDF.weibull <- pweibull(q, shape = 1.977, scale = 891.9)

survival.weibull <- 1 - CDF.weibull

# Create a survival object
surv_object <- Surv(DishwasherBreak$Cycles,
                    event = DishwasherBreak$Censoring.Indicator == "Failed" )

## Hazard function plot for the assumed distribution

hazard.fn <- psm(surv_object ~ 1, data = DishwasherBreak)

par(mfrow = c(2,2))

plot(x, PDF.weibull, type = "l", ylim = c(0, 0.001), xlab="Cycles")
plot(q, CDF.weibull, type = "l", xlab="Cycles")
plot(q, survival.weibull, type = "l", xlab="Cycles")

```

```

survplot(hazard.fn, what="hazard")

par(mfrow = c(1,1))

##ML Estimates tables

DishwasherBreak.mlest <- mlest(DishwasherBreak.ld,
                               distribution = "Weibull")

#print(DishwasherBreak.mlest)

print(DishwasherBreak.mlest$ll.text)
print(DishwasherBreak.mlest$mttf.text)

print(DishwasherBreak.mlest$mle.table)

print(DishwasherBreak.mlest$failure.probabilities)

print(DishwasherBreak.mlest$quantiles)

DishwasherObstruction <- read.csv("DishwasherReliabilityObstruction.csv")

## Failure Mode = Break, Censoring = Obstruction

##Lognormal fits better

DishwasherObstruction.ld <- frame.to.ld(DishwasherObstruction,
                                         response.column = 2,
                                         censor.column = 3,
                                         time.units = "Cycles")

par(mfrow= c(1,2))
mleprobplot(DishwasherObstruction.ld, distribution="Weibull")
legend("top", legend = "Mode 2 = Obstruction ~ Weibull" , lty = 1:1)

mleprobplot(DishwasherObstruction.ld, distribution="lognormal")
legend("top", legend = "Mode 2 = Obstruction ~ Lognormal" , lty = 1:1)
par(mfrow= c(1,1))

par(mfrow= c(1,2))
mleprobplot(DishwasherObstruction.ld, distribution="loglogistic")
legend("top", legend = "Mode 2 = Obstruction ~ Loglogistic" , lty = 1:1)

mleprobplot(DishwasherObstruction.ld, distribution="frechet")
legend("top", legend = "Mode 2 = Obstruction ~ Frechet" , lty = 1:1)
par(mfrow= c(1,1))

```

```

x<-7:1200

PDF.frechet <- dfrechet(x, scale = 1.95, shape = 4.928)

q <- qfrechet(x/1200,scale = 1.95, shape = 4.928)

CDF.frechet <- pfrechet(q, scale = 1.95, shape = 4.928)

survival.frechet <- 1-CDF.frechet

# Create a survival object
surv_object1 <- Surv(DishwasherObstruction$Cycles,
                     event = DishwasherObstruction$Censoring.Indicator == "Failed")

## Hazard function plot for the assumed distribution

hazard.fn1 <- psm(surv_object1 ~ 1, data = DishwasherObstruction)

par(mfrow = c(2,2))

plot(x, PDF.frechet, type = "l", ylim = c(0, 0.0015), xlab="Cycles")
plot(q, CDF.frechet, type = "l", xlab="Cycles")
plot(q, survival.frechet, type = "l", xlab="Cycles")
survplot(hazard.fn1, what="hazard")

par(mfrow = c(1,1))

DishwasherObstruction.mlest <- mlest(DishwasherObstruction.ld,
                                   distribution = "frechet")

#print(DishwasherObstruction.mlest)

print(DishwasherObstruction.mlest$ll.text)
print(DishwasherObstruction.mlest$mttf.text)

print(DishwasherObstruction.mlest$mle.table)

print(DishwasherObstruction.mlest$failure.probabilities)

print(DishwasherObstruction.mlest$quantiles)

mleprobplot(DishwasherBreak.ld, distribution="Weibull", lwd.ci = NULL)

```

```

mleprobplot(DishwasherObstruction.ld, distribution="frechet",
             param.loc = "topleft", lwd.ci = NULL, add = TRUE)

legend("top", legend = c("Mode 2 = Obstruction ~ Frechet",
                          "Mode 1 = Break ~ Weibull" ), lty = 1:1)

DishwasherBreak1 <- read.csv("DishwasherReliabilityBreak2.csv")

ti <- DishwasherBreak1$Cycles
di <- DishwasherBreak1$Failed
ri <- 0
n <- sum(DishwasherBreak1$Failed)
ni <- DishwasherBreak1$Risked
pi.hat<- di/ni
S.hat<- cumprod(1-pi.hat)
F.hat <- 1 - S.hat

DishwasherBreak.1 <- cbind(DishwasherBreak1, DishwasherBreak1$Cycles, S.hat, F.hat)

# Filter rows where Failure is "Break" and select S.hat and F.hat
result <- DishwasherBreak.1[DishwasherBreak1$Failure == "Break", c("Failure","Cycles","S.hat", "F.hat")]

# View the resulting data frame
print(result)

subset_break <- DishwasherBreak.1[DishwasherBreak1$Failure == "Break", ]

lab.vec <- c(0.0001,0.0002,0.0003,0.0005,0.001,0.003,0.005,0.01,0.02,0.03,
             0.05,0.1,0.2,0.3, 0.5, 0.8, 0.9, 0.99, 0.999)

plot(subset_break$Cycles, log(-log(1 - subset_break$F.hat)),
     ylab = "Probability", xlab = "Cycles",
     main = "Weibull Probability Scale",
     ylim = log(-log(1 - c(0.001, 0.999))), log = 'x', axes = FALSE,
     xlim = c(98, 2000))

# Add axes
axis(1)
axis(2, las = 2, at = log(-log(1 - lab.vec)), tcl = 0.1, labels = lab.vec)

# Create a survival object
surv_object <- Surv(DishwasherBreak1$Cycles,
                    event = DishwasherBreak1$Censoring.Indicator == "Failed")

```

```

survfit_object <- survfit(surv_object ~ 1, data = DishwasherBreak1)

summary_result<- summary(survfit_object, time = DishwasherBreak1$Cycles)

# Create a data frame
summary_df <- data.frame(time = summary_result$time,
                          survival = summary_result$surv,
                          n.risk = summary_result$n.risk,
                          n.event = summary_result$n.event,
                          std.err = summary_result$std.err,
                          "lower 95 CI" = summary_result$lower,
                          "upper 95 CI" = summary_result$upper)

summary_df_break <- subset(summary_df, n.event == 1)

print(summary_df_break)

# Plot the Kaplan-Meier survival curve

# Create a survival object
surv_object.1 <- Surv(subset_break$Cycles,
                      event = subset_break$Censoring.Indicator == "Failed")

survfit_object.1 <- survfit(surv_object.1 ~ 1, data = subset_break)

plot(survfit_object.1, col = 1, mark.time = TRUE, main = "Kaplan-Meier Survival Curve",
     xlab = "Time to Event", ylab = "Survival Probability")

DishwasherObstruction1 <- read.csv("DishwasherReliabilityObstruction2.csv")

#plot the hazard cumulative function for failure mode = break

ti <- DishwasherObstruction1$Cycles
di <- DishwasherObstruction1$Failed
n <- sum(DishwasherObstruction1$Failed)
ni <- DishwasherObstruction1$Risked
pi.hat<- di/ni
S.hat1<- cumprod(1-pi.hat)
F.hat1 <- 1 - S.hat1

```

```

DishwasherObstruction.1 <- cbind(DishwasherObstruction1,DishwasherObstruction1$Cycles, S.hat1, F.hat1)

# Filter rows where Failure is "Break" and select S.hat and F.hat
result.o <- DishwasherObstruction.1[DishwasherObstruction1$Failure == "Obstruction",
c("Failure","Cycles","S.hat1", "F.hat1")]

# View the resulting data frame
print(result.o)

subset_obst <- DishwasherObstruction.1[DishwasherObstruction1$Failure == "Obstruction", ]

lab.vec <- c(0.0001,0.0002,0.0003,0.0005,0.001,0.003,0.005,0.01,0.02,0.03,
0.05,0.1,0.2,0.3, 0.5, 0.8, 0.9, 0.99, 0.999)

plot(subset_obst$Cycles, log(-log(1 - subset_obst$F.hat1)),
      ylab = "Probability", xlab = "Cycles",
      main = "Weibull Probability Scale",
      ylim = log(-log(1 - c(0.001, 0.999))), log = 'x', axes = FALSE,
      xlim = c(7, 1500))

# Add axes
axis(1)
axis(2, las = 2, at = log(-log(1 - lab.vec)), tcl = 0.1, labels = lab.vec)

# Create a survival object
surv_object1 <- Surv(DishwasherObstruction1$Cycles,
                     event = DishwasherObstruction1$Censoring.Indicator == "Failed")

survfit_object1 <- survfit(surv_object1 ~ 1, data = DishwasherObstruction1)

summary_result1<- summary(survfit_object1, time = DishwasherObstruction1$Cycles)

# Create a data frame
summary_df1 <- data.frame(time = summary_result1$time,
                          survival = summary_result1$surv,
                          n.risk = summary_result1$n.risk,
                          n.event = summary_result1$n.event,
                          std.err = summary_result1$std.err,
                          "lower 95 CI" = summary_result1$lower,
                          "upper 95 CI" = summary_result1$upper)

summary_df_obst <- subset(summary_df1, n.event == 1)

print(summary_df_obst)

```

```

# Create a survival object
surv_object.2 <- Surv(subset_obst$Cycles,
                      event = subset_obst$Censoring.Indicator == "Failed")

survfit_object.2 <- survfit(surv_object.2 ~ 1, data = subset_obst)

# Plot the Kaplan-Meier survival curve

plot(survfit_object.2, col = 1, mark.time = TRUE, main = "Kaplan-Meier Survival Curve",
     xlab = "Time to Event", ylab = "Survival Probability")

Dishwasher2 <- read.csv("DishwasherReliability2.csv")

eta.w <- 891.9
beta.w <- 1.977

mu.f <- 4.928
sigma.f <- 1.95

eta.f <- exp(mu.f)
beta.f <- 1/sigma.f

F.hat.joint <- 1 - exp(-1*(Dishwasher2$Cycles/eta.w)^beta.w) +
  exp(-1*((Dishwasher2$Cycles/eta.w)^beta.w + (eta.f/Dishwasher2$Cycles)^beta.f))

S.hat.joint <- 1- F.hat.joint

Cycles <- Dishwasher2$Cycles

cbind(Cycles, F.hat.joint, S.hat.joint)

Dishwasher2 <- read.csv("DishwasherReliability2.csv")
# Kaplan - Meier Estimate

ti <- Dishwasher2$Cycles
di <- Dishwasher2$Failed
n <- sum(Dishwasher2$Failed)
ni <- Dishwasher2$Riskd
pi.hat<- di/ni
S.hat.KM<- cumprod(1-pi.hat)
F.hat.KM <- 1 - S.hat.KM

Cycles <- Dishwasher2$Cycles

Dishwasher.2 <-cbind(Cycles,F.hat.KM, S.hat.KM)
print(Dishwasher.2)

```

```

### Plotting the Kaplan - Meier estimate

lab.vec <- c(0.0001,0.0002,0.0003,0.0005,0.001,0.003,0.005,0.01,0.02,0.03,
            0.05,0.1,0.2,0.3, 0.5, 0.8, 0.9, 0.99, 0.999)

plot(Cycles, log(-log(1 - F.hat.KM)),
     ylab = "Probability", xlab = "Cycles",
     main = "Weibull Probability Scale",
     ylim = log(-log(1 - c(0.001, 0.999))), log = 'x', axes = FALSE,
     xlim = c(7, 2000))

# Add axes
axis(1)
axis(2, las = 2, at = log(-log(1 - lab.vec)), tcl = 0.1, labels = lab.vec)

Dishwasher <- read.csv("DishwasherReliability1.csv")

surv_object3 <- Surv( time = Dishwasher$Cycles
                     ,time2 = Dishwasher$Cycles
                     ,type = "interval2")

#Fitting Separate Distributions to each failure mode
SepDist.fits <- lapply(c("Break", "Obstruction"),
                      function(failure)
                        survreg(Surv( time = Cycles
                                      ,time2 = Cycles
                                      ,type = "interval2") ~ 1,
                                data=subset(Dishwasher, Dishwasher$Failure == failure),
                                dist = "weibull"))

print(SepDist.fits)

log.l.break <- -184.1

log.l.obstruction <- -243.9

log.l.sep <- log.l.break + log.l.obstruction

print(log.l.sep)

log_likelihood.break <- function(parameters, data) {
  shape <- parameters[1]
  scale <- parameters[2]

  # Ensure shape and scale are positive
  if (shape <= 0 || scale <= 0) {

```

```

    return(Inf) # Return infinite log-likelihood for invalid parameters
  }

  log_likelihood_values <- dweibull(data, shape = shape, scale = scale, log = TRUE)
  return(-sum(log_likelihood_values))
}

DishwasherBreakonly<- read.csv("DishwasherBreakonly.csv")
cycles.break <- DishwasherBreakonly$Cycles

# Initial parameter values
initial_params <- c(shape = 1, scale = 1)

# Optimize the log-likelihood function
result.break <- optim(par = initial_params, fn = log_likelihood.break, data = cycles.break)

# Extract the optimized parameters
optimized_params.break <- result.break$par

# View optimization results
print(result.break)
print(optimized_params.break)

log_likelihood.obstr <- function(parameters, data) {
  shape <- parameters[1]
  scale <- parameters[2]

  # Ensure shape and scale are positive
  if (shape <= 0 || scale <= 0) {
    return(Inf) # Return infinite log-likelihood for invalid parameters
  }

  log_likelihood_values <- dweibull(data, shape = shape, scale = scale, log = TRUE)
  return(-sum(log_likelihood_values))
}

# Example dataset
set.seed(123)

DishwasherObstructiononly<- read.csv("DishwasherObstructiononly.csv")
cycles.obstr <- DishwasherObstructiononly$Cycles

# Initial parameter values
initial_params <- c(shape = 1, scale = 1)

# Optimize the log-likelihood function
result.obstr <- optim(par = initial_params, fn = log_likelihood.obstr, data = cycles.obstr)

# Extract the optimized parameters
optimized_params.obstr <- result.obstr$par

```

```

# View optimization results
print(result.obstr)
print(optimized_params.obstr)

#Assuming both failure modes have equal sigma

Equal.fit <- survreg(surv_object3 ~ as.factor(Failure),
                     data = Dishwasher,
                     dist="weibull")

print(Equal.fit)

log.l.equal <- -433.1

print(LRtest <- 2*(log.l.sep -log.l.equal))

print(c.value <- qchisq(0.95, 1))

```